



Contents lists available at ScienceDirect

Journal of Sound and Vibration

journal homepage: www.elsevier.com/locate/jsv

Micro-vibration model and parameter estimation method of a reaction wheel assembly

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ARTICLE INFO

Article history:

Received 18 August 2013

Received in revised form

24 February 2014

Accepted 13 April 2014

Handling Editor: L.G. Tham

ABSTRACT

Reaction wheel assemblies (RWAs) are a source of disturbance in satellites, and they are regarded as the largest jitter contributor in optical payloads. In order to ensure a stringent jitter requirement, the wheel disturbance effects on spacecraft should be predicted precisely prior to launch through analytical or experimental approaches. For this purpose, the wheel disturbance should be identified and modeled accurately. In the present study, a micro-vibration model of the RWA is introduced through coupling an analytical wheel model and an empirical disturbance model; furthermore, a parameter estimation process of the coupled model from the micro-vibration disturbance data is proposed. In order to verify the modeling and estimation techniques, a micro-vibration model of a numerical RWA is established and its estimation error is validated. Then, the micro-vibration model is extended to consider an axial disturbance and a measurement offset effect. Finally, the micro-vibration model is applied to a commercial RWA and the model parameters are extracted from the disturbance test data of the RWA using the parameter estimation process. The analytical and experimental results demonstrate that the proposed micro-vibration model and parameter estimation process are effective in the dynamic disturbance modeling of RWAs.

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1. Introduction

These days, space observation satellites have very precise optical instruments that are used for various purposes and delicate space-based missions. The optical payloads are very sensitive to high frequency pointing disturbances and, thus, with small amplitudes, the dynamic perturbations (micro-vibrations) on optical mirrors and detectors can cause significant effects in the image quality degradation, such as optical path length differences, blurring, and smears in the images [1,2]. In order to ensure mission success, spacecraft requirements include having extremely high pointing accuracy, stability, and sub-arcsecond level line-of-sight (LOS) jitter [3–5]. Moreover, compliance with the jitter requirements should be verified prior to launch through analytical and/or experimental approaches [6–8]. Therefore, accurate dynamic models of the disturbances should be established in order to demonstrate a good estimation of the jitter performance.

The dynamic disturbances of satellites are induced either by external sources in the space environment or by internal sources from the actuator and sensor systems onboard the spacecraft [9]. Micro-vibrations are primarily generated by the onboard mechanical moving systems including mechanically tunable optical filters, crycoolers, solar array drive

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<http://dx.doi.org/10.1016/j.jsv.2014.04.032>

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mechanisms, antenna pointing mechanisms, and reaction wheel assemblies (RWAs). Among these micro-vibration disturbance sources, the disturbance produced by RWAs is generally regarded as the largest [10]. The RWA is a momentum exchange device that is used in spacecraft attitude stabilization and control, and it provides axial reaction torques and stores angular momentums through adjusting the rotational speed of the flywheel. A typical RWA, as depicted in Fig. 1, consists of a rotating flywheel suspended on ball or magnetic bearings, and is driven by an internal brushless DC motor [11,12].

The micro-vibration disturbances induced by the RWA are generally categorized into fundamental and sub and/or higher harmonic disturbances. The fundamental harmonic disturbances are considered to be the most significant micro-vibration source; they are caused by the static and dynamic imbalances of the flywheel and it produces a disturbance force and moment according to the wheel speed. The sub/higher harmonic disturbances are caused by the dynamic lubrication behavior, motor disturbance, control error (such as cogging and torque ripple), and bearing irregularities in their balls, races, and cage. This harmonic disturbance also produces a disturbance force and moment at sub/higher harmonics of the wheel speed [13]. In addition, RWAs have structural vibration modes as a result of the internal flexibility of the wheel structure. As shown in Fig. 2, there are three dominant structural modes: lateral mode, axial mode, and rocking (whirl) mode [14]. The harmonic disturbances can be amplified by these vibration modes when the harmonics cross the mode frequency. In some cases, the resonant peaks of the amplified disturbances, as measured in this study, can be significantly greater than the fundamental harmonic disturbance, which results in a significant jitter amplitude. Therefore, in order to obtain an accurate disturbance model, the dynamic flexibility of the RWAs must also be considered in the modeling process.

Over the past few decades, there have been numerous studies on RWA disturbance modeling. The disturbance modeling methods can be broadly classified into empirical models and analytical models. The empirical models use the stochastic broadband model or the geometric relationship of the wheel to establish several discrete harmonic disturbances [12,15–19]. Masterson et al. [15,16] applied a MATLAB toolbox in order to extract the disturbance coefficients of an empirical disturbance model from the RWA micro-vibration data after removing the structural resonant data. Taniwaki and Ohkami [17] developed a new model that detects low-frequency disturbances using the wheel geometric relationship and contact dynamics. Oh and Cheon [18] used an empirical model to suggest a calibration process that reduces the resonance frequency error of a measurement table. Zhao et al. [19] suggested an identification method for the amplitude coefficients of an empirical model using the energy compensation method. These empirical models are very useful in representing the RWA disturbances in discrete harmonic forms; however, they cannot capture the resonance effect between the harmonic

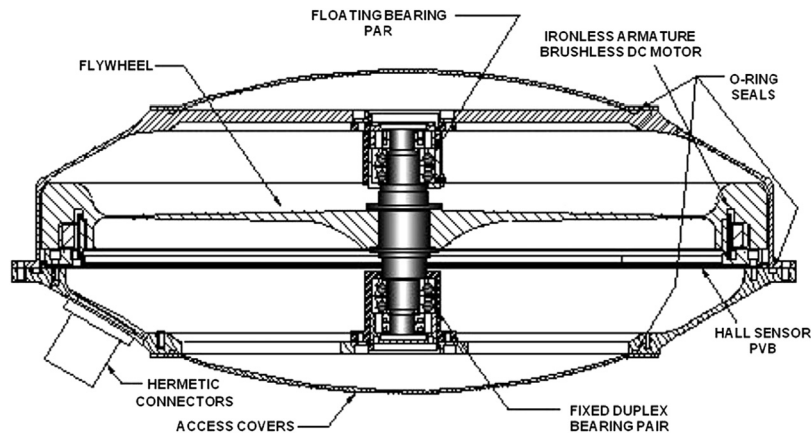


Fig. 1. Cross section of ITHACO E-type reaction [12].

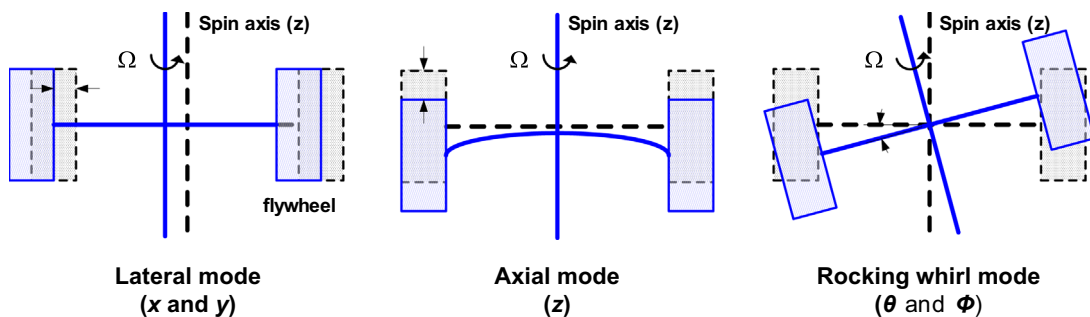


Fig. 2. Structural modes of a reaction flywheel.

disturbances and the RWA structural modes, and thus, they do not necessarily guarantee the estimation accuracy of the harmonic coefficients.

The analytical models use physical dynamic models of wheels, which have been established using the energy method (Lagrange's equation) or the Newton–Euler method, in order to capture the internal flexibility of the wheel [8,15,20–23]. Masterson et al. [15] developed an analytical model using the energy method that considered the wheel dynamic modes, except the axial translation motion, and combined the analytical model with its empirical model, where the transmitted disturbances were represented only using the spring stiffness; then, Liu et al. [8] used the analytical model to predict the jitter response induced from an RWA. Using the energy method, Zhang et al. [20,21] established an analytical model for a cantilevered RWA supported by a soft suspension system, which was designed as a passive control system in order to minimize the high frequency disturbances. Zhou et al. [22,23] used the Newton–Euler method to develop an analytical model for a dynamic system that consisted of an intelligent isolation platform and a wheel assembly. These analytical models can be extended for application in coupling analyses between a spacecraft structure and the RWAs [24–26]. However, the analytical models have numerous parameters that should be determined accurately, and a methodology to determine the parameter values using experimental vibration data has not yet been discussed in detail.

In this study, a practical and systematic estimation approach that can accurately extract the parameter values of the RWA and disturbance models from the vibration test data is proposed. The transmitted forces and moments are represented using the damping coefficient as well as the spring stiffness, so that the estimation accuracy of the disturbance and damping coefficients can be improved. In addition, the analytical wheel model is expanded in the z-axis in order to capture the axial mode. Moreover, vertical and horizontal offsets are added to the micro-vibration model in order to consider the coupling effect between the transmitted forces and moments. The modeling and estimation methods proposed in this study are validated using numerical and experimental examples. The organization of this paper is as follows. In the next section, the micro-vibration model of the RWA is described briefly based on the coupling between the analytical wheel model and the empirical disturbance model, and the parameter estimation method of the micro-vibration model is proposed. In order to verify the modeling and estimation methods, the coupled model of a numerical RWA is established and the modeling accuracy is confirmed through evaluating the parameter errors. The micro-vibration test is performed on a commercial RWA, and the micro-vibration model is improved in order to consider an axial disturbance and a measurement offset effect. Finally, the modeling and estimation methods are applied to the RWA and the modeling error and vertical offset effect are investigated.

2. Micro-vibration model

The micro-vibration model for RWAs consists of a structural dynamic model (analytical model) and a disturbance model (empirical model). In this section, the equations of motion for the RWA and its harmonic disturbance model are summarized. Then, a parameter estimation method for determining the parameter values of the micro-vibration model is introduced.

2.1. Reaction wheel model

The dynamic part of a conventional RWA has a symmetric configuration based on the flywheel. The degree of freedom (dof) can be expressed using a reference frame, as depicted in Fig. 3, which is fixed at the center of gravity (CoG) of the flywheel and is parallel to the wheel rotation axis in the z-axis. The supporting parts, such as the flexible shaft and the upper/lower bearings, can be modeled using the linear springs and dashpots at each axis. The static imbalance (U_s) is caused by an offset of the center of mass of the flywheel from the rotating axis. This produces a rotating, radial force disturbance at the wheel speed and can be generally represented by a small mass (m_s) at the radius. The dynamic imbalance (U_d) is caused by an angular misalignment between the principle axis of the wheel and the spin axis: it produces a rotating, radial moment disturbance at the wheel speed and can be represented using two equal masses (m_d) [15]. These imbalances can be expressed as follows:

$$U_s = m_s r_s, \quad (1)$$

$$U_d = 2m_d r_d h, \quad (2)$$

where r_s and r_d are the radial distances of the static and dynamic imbalance masses, respectively, and h is the axial distance of the dynamic imbalance mass from the center of the flywheel.

For simplicity, in this modeling, it is assumed that the wheel is free along the spin axis (z-axis) and has four dofs: two radial translations (x and y) and two radial rotations (θ and ϕ). Using the energy method and the assumption of symmetrical material properties, the equations of motion can be expressed as follows:

$$\begin{bmatrix} m_t & 0 \\ 0 & m_t \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{y} \end{Bmatrix} + \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix} + \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} F_x \\ F_y \end{Bmatrix} \quad (3)$$

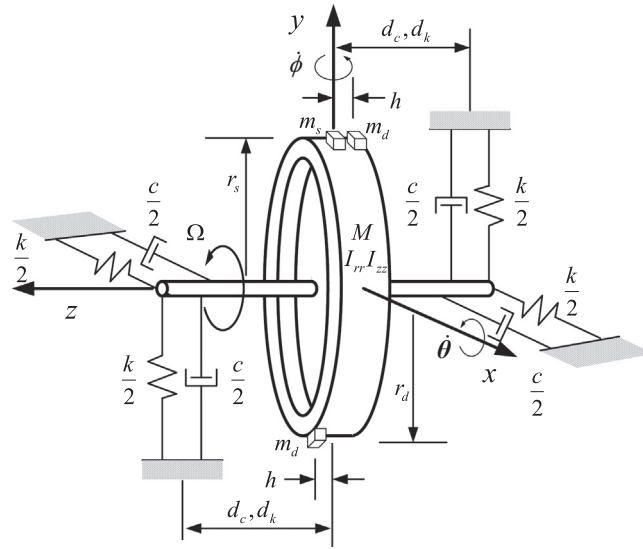


Fig. 3. Analytical reaction wheel assembly model: modified from [15].

$$\begin{bmatrix} I_{rr} & \frac{1}{2}\tilde{I} \sin(2\Omega t) \\ \frac{1}{2}\tilde{I} \sin(2\Omega t) & I_{rr} \end{bmatrix} \begin{Bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{Bmatrix} + \Omega \begin{bmatrix} -\tilde{I} \sin(2\Omega t) & I_{zz} \\ -I_{zz} & \tilde{I} \sin(2\Omega t) \end{bmatrix} \begin{Bmatrix} \dot{\theta} \\ \dot{\phi} \end{Bmatrix} + \begin{bmatrix} c_\theta & 0 \\ 0 & c_\phi \end{bmatrix} \begin{Bmatrix} \dot{\theta} \\ \dot{\phi} \end{Bmatrix} + \begin{bmatrix} k_\theta & 0 \\ 0 & k_\phi \end{bmatrix} \begin{Bmatrix} \theta \\ \phi \end{Bmatrix} = \begin{Bmatrix} T_x \\ T_y \end{Bmatrix}, \quad (4)$$

where Ω is the rotating speed of the wheel; c and k are the damping coefficient and stiffness of the supporting structure in the translation axes, respectively; c_θ and k_θ are the corresponding damping coefficient and stiffness in the rotation axes; I_{rr} and I_{zz} are the radial and polar moments of inertia (MOI) of the balanced flywheel, respectively; $\{F_x, F_y\}^T$ and $\{T_x, T_y\}^T$ are the disturbance force and moment vectors, respectively.

The total mass (m_t) and total MOI ($\tilde{I}$), including the static and dynamic imbalance masses, can be written as follows:

$$m_t = M + m_s + 2m_d, \quad (5)$$

$$\tilde{I} = 2m_d r_d^2 + m_s r_s^2, \quad (6)$$

where M is the flywheel mass.

The wheel dynamic structure has two dominant modes: the lateral mode and the rocking whirl mode. By solving the eigenvalue problems in Eqs. (2) and (3) with an assumption that the wheel is an undamped structure, the frequencies of the lateral and whirl modes can be obtained as follows:

$$\omega^{\text{lateral}} = \sqrt{\frac{k}{m_t}}, \quad (7)$$

$$\omega_{1,2}^{\text{whirl}} = \mp \frac{\Omega I_{zz}}{2I_{rr}} + \sqrt{\left(\frac{\Omega I_{zz}}{2I_{rr}}\right)^2 + \frac{k_\theta}{I_{rr}}}, \quad (8)$$

where $\omega_{1,2}^{\text{whirl}}$ represents the negative and positive whirl modes.

2.2. Disturbance model

The disturbance induced during the wheel operation is an internal perturbation that consists of fundamental harmonic disturbances and sub/higher harmonic disturbances. The fundamental harmonic disturbance force and moment are caused by the static and dynamic imbalances expressed in Eqs. (1) and (2). The sub/higher harmonics are caused by various disturbance sources such as bearing irregularities, dynamic lubrication behavior, and motor disturbances. These disturbing harmonics occur not only at integer multiples of the wheel speed but also at non-integer multiples of the fundamental frequency. Using empirical models [15], all fundamental and higher harmonic disturbances can be expressed only using the

discrete harmonics of the reaction wheel speed with amplitudes proportional to the wheel speed squared, as given below:

$$\begin{aligned} F_x &= - \sum_{i=1}^{n_f} C_i^f \Omega^2 \sin(h_i^f \Omega t + \phi_i^f) \\ F_y &= \sum_{i=1}^{n_f} C_i^f \Omega^2 \cos(h_i^f \Omega t + \phi_i^f) \\ T_x &= \sum_{i=1}^{n_t} C_i^t \Omega^2 \cos(h_i^t \Omega t + \phi_i^t) \\ T_y &= \sum_{i=1}^{n_t} C_i^t \Omega^2 \sin(h_i^t \Omega t + \phi_i^t) \end{aligned} \quad (9)$$

where C_i is the harmonic amplitude of the i th harmonic, h_i is the i th harmonic number, and ϕ_i is a random phase angle.

Unfortunately, it is not possible to directly measure the wheel disturbances expressed in Eq. (9) using practical vibration tests on RWAs. Once the wheel disturbances are generated by the wheel operation, the disturbances are transmitted to an adjacent structure, e.g. a wheel bracket or measurement device, through the internal flexibility of the wheel. Therefore, the micro-vibration data obtained by a measurement system such as a Kistler table are the transmitted disturbances. As shown in Fig. 3, the rotating flywheel is supported by the springs and dampers; thus, the transmitted forces ($\bar{F}$) and moments ($\bar{T}$) can be expressed using the structural responses as follows:

$$\begin{Bmatrix} \bar{F}_x \\ \bar{F}_y \\ \bar{T}_x \\ \bar{T}_y \end{Bmatrix} = \begin{Bmatrix} c\dot{x} + kx \\ c\dot{y} + ky \\ c_\theta\dot{\theta} + k_\theta\theta \\ c_\theta\dot{\phi} + k_\theta\phi \end{Bmatrix}. \quad (10)$$

2.3. Parameter estimation method

The micro-vibration model of the RWA can be established by coupling the reaction wheel model expressed in Eqs. (3) and (4), and the disturbance model in Eq. (9). The validity of the micro-vibration model depends on the accuracy of the parameter values in the models. In the present study, a parameter estimation method is proposed for the extraction of the model parameters from the RWA transmitted disturbance data. The proposed estimation method is a systematic approach based on reasonable assumptions; the assumption and estimation processes are proposed as follows.

2.3.1. Model assumptions

- The static and dynamic imbalance masses are sufficiently small to be negligible when compared with the flywheel mass ($m_s, m_d \ll M$).
- The moment of inertia caused by the imbalance masses is sufficiently small to be negligible when compared with the radial and polar moments of inertia ($\bar{I} \ll I_{rr}, I_{zz}$).
- The mass (M) and polar MOI (I_{zz}) of the flywheel are scalable mechanical properties, and these values are already known (typically provided by the vendor).

2.3.2. Parameter estimation process

- Step 1: Determine the main harmonic orders (h_i^f and h_i^t) from the frequency spectrum of the transmitted disturbance data.
- Step 2: Estimate the lateral stiffness (k) using a nonlinear optimization process of the lateral mode frequency function as expressed in Eq. (7) with the lateral mode frequency data through minimizing the estimation error.
- Step 3: Estimate the torsional stiffness (k_θ) and radial MOI (I_{rr}) using the optimization process of the whirl mode frequency function as expressed in Eq. (8) with the whirl mode frequency data.
- Step 4: Estimate the damping coefficients (c and c_θ) and harmonic amplitudes (C_i^f and C_i^t) using the optimization process through minimizing a cost function. Taking the Laplace transform of Eqs. (3), (4), (9), and (10), and then coupling those equations, the cost function of the i th harmonic can be obtained as follows:

$$f_i^{\text{cost}} = \sum_{j \in S_i} \left\{ \sum_{\omega_1}^{\omega_2} \left| \bar{\mathbf{F}}_i(\omega) - \left(\frac{\mathbf{C} \cdot \mathbf{j}\omega + \mathbf{K}_{\text{total}}}{-\mathbf{M}_{\text{total}} \cdot \omega^2 + \mathbf{C}_{\text{total}} \cdot \mathbf{j}\omega + \mathbf{K}_{\text{total}}} \cdot \tilde{\mathbf{C}}_i \right) \cdot \Omega^2 \right| \right\} \rightarrow 0; \quad 1 \leq S_i \leq 4, \quad (11)$$

where i is the harmonic order, j is the degree of freedom (dof) that belongs to the i th harmonic dof set (S_i), ω is the harmonic frequency ($\omega = i\Omega$), ω_1 and ω_2 are the start and end harmonic frequencies, respectively, $\mathbf{C}$ is the third part of Eq. (4), $\tilde{\mathbf{C}}_i$ is the estimated harmonic amplitude vector of the i th harmonic, $\bar{\mathbf{F}}_i(\omega)$ is the transmitted disturbance vector of

the i th harmonic at ω , and $\mathbf{M}_{\text{total}}$, $\mathbf{C}_{\text{total}}$, and $\mathbf{K}_{\text{total}}$ are the total mass, damping, and stiffness matrices of Eqs. (3) and (4), respectively.

This parameter estimation is conducted using two sequential optimization processes. From the first optimization performed for both damping coefficients and harmonic amplitudes, the damping coefficients (c and c_θ) are estimated differently depending on the harmonic order; then, their final optimized values are determined through averaging the estimated values. Finally, the optimized values of the harmonic amplitudes (C_i^f and C_i^t) are determined through the second optimization that is only performed on the harmonic amplitudes using the mean values of the damping coefficients.

3. Numerical example

A numerical analysis was performed in order to validate the micro-vibration model and the parameter estimation process described in the previous section. First, the dynamic response of a numerical reaction wheel was calculated through a coupling analysis between Eqs. (3), (4), and (9) using predefined parameter values; then, the transmitted disturbance forces and moments of the wheel were obtained from Eq. (10). Finally, the parameter estimation process was followed using the transmitted micro-vibration data, and the estimation error was verified through comparison with the input parameter values in order to verify the accuracy of the estimation process.

3.1. Reaction wheel model

In this example, it is assumed that the numerical reaction wheel has four dofs, i.e. two translations (x and y) and two rotations (θ and ϕ), that the dynamic response of the wheel model can be completely reconstructed using the equations of motion expressed in Eqs. (3) and (4), and that all the parameter values of the micro-vibration model are given precisely as listed in Table 1.

3.2. Disturbance model

In order to reconstruct the micro-vibration disturbances of the numerical wheel, the parameter values of the disturbance model expressed in Eq. (9) should be determined. In this example, the harmonic orders and amplitudes measured in Ref. [12] were used to generate the radial forces and moments; the harmonic disturbances were assumed to be same in both radial directions. The input disturbance parameter values are listed in Table 2, where the coefficient units are 10^{-7} N/rpm² for forces and 10^{-7} Nm/rpm² for moments.

In this numerical simulation, the wheel speed was increased from 0 rpm to 2500 rpm, and the accumulated cycle function [27] was used to create the sine wave signal that represents the continuous variation of the wheel speed. The wheel speed and radial disturbances are shown in Fig. 4. The waterfall plots of the radial force and moment, as shown in Fig. 5, clearly exhibit the discrete harmonics with amplitudes that are proportional to the wheel rate squared.

The structural response of the wheel can be obtained through the dynamic analysis of the wheel model for the harmonic disturbances, and then the transmitted disturbances can be calculated using Eq. (10). The waterfall plots of the transmitted disturbances are shown in Fig. 6, which clearly demonstrates the amplification of the harmonic disturbances as a result of the wheel flexibility effect.

Table 1
Parameter values of the numerical reaction wheel model.

Parameters	Values
Lateral mode freq.	150 Hz
Whirl mode freq.	100 Hz
Flywheel mass (M)	5.9 kg
Polar MOI (I_{zz})	0.075 kg m ²
Radial MOI (I_{rr})	0.063 kg m ²
Imbalance radial distance (r_s, r_d)	0.16 m
Dynamic imbalance axial distance (h)	0.048 m
Static imbalance (U_s)	0.716 g cm
Dynamic imbalance (U_d)	29.54 g cm ²
Lateral stiffness (k)	5.241×10^6 N/m
Lateral damping coeff. (c)	200 N s/m
Rotational stiffness (k_θ)	2.467×10^4 Nm/rad
Rotational damping coeff. (c_θ)	0.942 Nm s/rad

Table 2
Harmonic orders and coefficients of the numerical disturbance model.

Harmonic order (i)	Radial force (F_x, F_y)		Radial moment (C_x, C_y)	
	h_i^f	C_i^f	h_i^c	C_i^c
1	1.00	0.7852	1.00	0.3239
2	2.00	0.2984	2.00	0.0744
3	/	/	2.90	0.0309
4	3.00	0.0777	/	/
5	/	/	3.88	0.0472
6	4.00	0.1036	4.00	0.0121
7	4.43	0.1280	4.43	0.0412
8	5.00	0.0849	/	/
9	/	/	5.20	0.0924
10	5.38	0.0985	/	/
11	/	/	5.40	0.1027
12	5.60	0.1849	5.60	0.0630
13	/	/	5.81	0.0851
14	6.00	0.0618	6.00	0.0173
15	6.39	0.0402	/	/

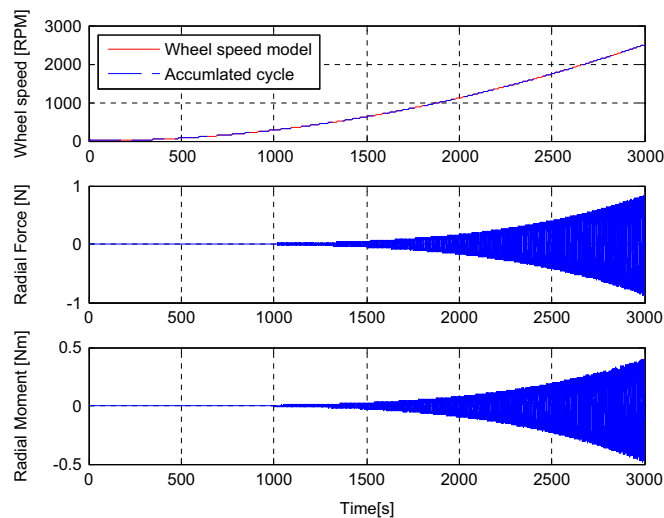


Fig. 4. Time histories of wheel speed and disturbances: wheel speed=0–2500 rpm.

3.3. Model estimation and verification

The parameter estimation method suggested in this study was applied to the transmitted disturbance data. In this estimation process, as mentioned previously, it was assumed that the wheel mass and polar MOI are the known values, and the static and dynamic imbalance masses were ignorable. In the estimation process, the optimum parameter values were determined using the MATLAB[®] Optimization Toolbox ('fminsearch'), which is generally referred to as an unconstrained nonlinear optimization, and using the Nelder–Mead simplex algorithm [28]. In this parameter optimization, the termination tolerances for the function value and variable ('TolFun' and 'TolX', respectively) were set to 10^{-8} . The parameter estimation process was performed as follows.

Step 1: Harmonic order

The disturbance harmonic orders (h_i^f and h_i^c) were determined using the disturbance waterfall data through the harmonic identification method [16]. The harmonic order results are shown in Fig. 7, and the estimation error was less than 0.12 percent for both radial force and moment disturbances.

Step 2: Lateral stiffness

Using Eq. (7), the flywheel mass (M), and the lateral mode frequency data, the lateral stiffness (k) was estimated and the estimation error was 0.26 percent. Fig. 8 (upper plot) clearly exhibits the good agreement of the wheel model with the mode frequency data.

Step 3: Torsional stiffness and polar MOI

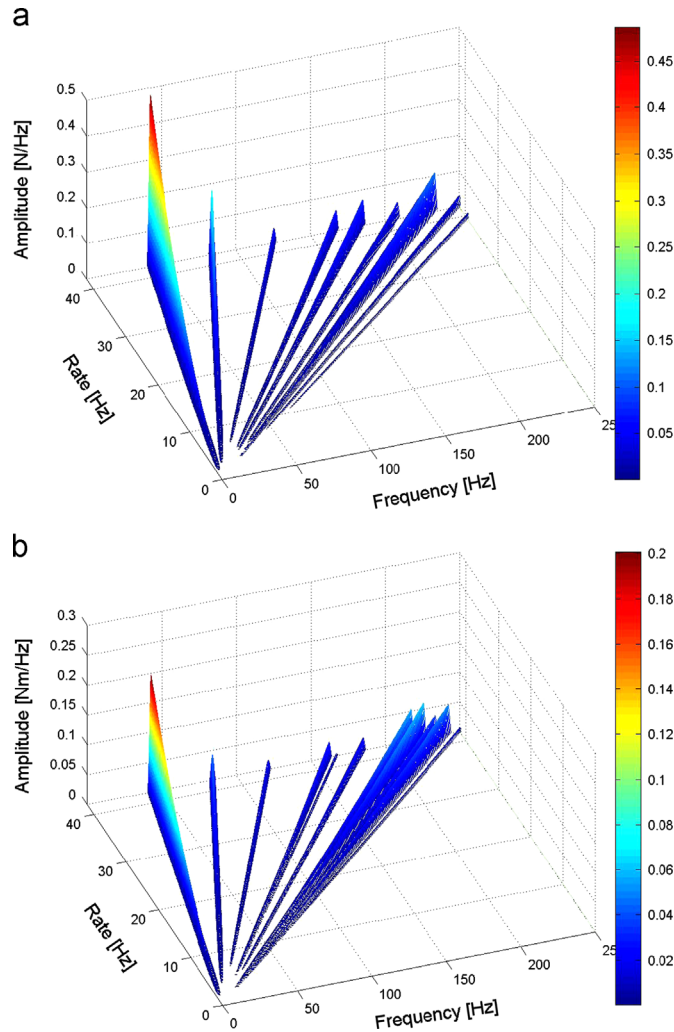


Fig. 5. Waterfall plots of input disturbances. (a) Radial force and (b) radial moment.

The torsional stiffness (k_θ) and radial MOI (I_{rr}) were estimated using Eq. (8) and the whirl mode frequency data, and the estimation errors of k_θ and I_{rr} were 0.27 percent and 0.36 percent, respectively. Fig. 8 (lower plot) clearly exhibits the good agreement of the wheel model with the negative and positive whirl frequency data.

Step 4: Damping and disturbance coefficients

Through the nonlinear parameter optimization process, the damping coefficients (c and c_θ) and the harmonic amplitudes (C_i^f and C_i^t) were estimated using the cost function expressed in Eq. (11). The estimation errors of c and c_θ were 0.51 percent and 1.18 percent, respectively, and the maximum errors of C_i^f and C_i^t were 0.56 percent and 2.43 percent, respectively. The estimated results of the micro-vibration model are shown in Fig. 9, where the three harmonics are selected for visibility.

This numerical simulation exhibits a much better fit than that accomplished in [12,15]. The discrepancy of the estimation results can be explained through the two significant distinctions of this method. The first distinction is that the transmitted disturbances were represented using both the spring stiffness (k) and damping coefficients (c), as described in Eq. (10); thus, the damping effect could be fully reflected in the cost function as described in Eq. (11). The second distinction is that the disturbance harmonic amplitudes (C_i) and damping coefficients (c) were estimated simultaneously in the fourth step of the parameter estimation process, so that both the width and height of the disturbance peaks could be captured.

From the analysis results, it is evident that the estimated results of the micro-vibration model agree very well with the transmitted harmonic disturbances and that the structural flexibility effect of the wheel can be well captured using the micro-vibration modeling method. Note also that the parameter values of the micro-vibration model can be accurately estimated using the proposed estimation process.

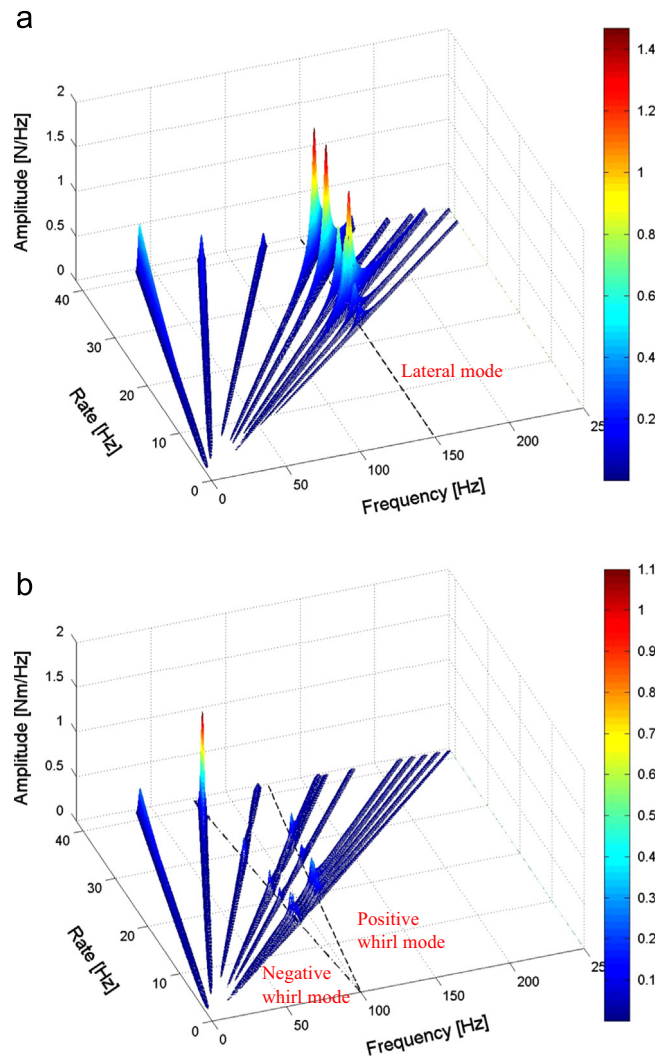


Fig. 6. Waterfall plots of transmitted disturbances. (a) Radial force and (b) radial moment.

4. Experimental application

The micro-vibration modeling method was applied to a commercial RWA. In order to measure the disturbances in the RWA, the micro-vibration test was performed using a high resolution dynamometer (Kistler table). For this application, the equation of the wheel motion was extended in order to consider the vertical translation motion of the wheel and the offset effect of the measurement system; then, the parameter estimation process was followed. The micro-vibration model was validated through comparing the micro-vibration model with the measured data.

4.1. Micro-vibration test

Because the internal disturbance source of RWAs cannot be measured directly, as mentioned above, the transmitted disturbance should be used in the micro-vibration modeling process. In this research, the micro-vibration test was performed on an engineering model (EM) of an ITHACO E-type wheel. Fig. 10 shows the configuration of the RWA micro-vibration measurement system. Using a rigid bracket, the EM wheel was mounted rigidly on a marble block supported by an air-floating isolation system installed on a seismic isolation floor. In the micro-vibration test, a quartz type dynamometer (Kistler/9255B) was used to simultaneously measure the six components of the disturbance (i.e. three forces and three moments). Fig. 11 shows the real configuration of the EM wheel model installed on the Kistler table.

In this test, a free run-down method was used to measure the micro-vibration disturbances continuously in the speed range from 1400 rpm to 0 rpm. After increasing the wheel speed to the maximum speed, the command input voltage to the RWA was set to zero so that the speed decreased continuously due to the friction torque of the flywheel. During the free run-down period, all test data were logged using a data acquisition system (dSAPCE/DS1103) at a sample rate of 2048 Hz through

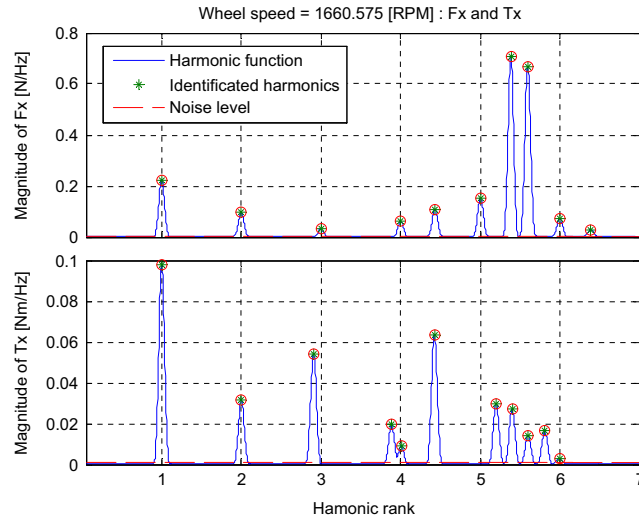


Fig. 7. Wheel harmonic identification results of radial force (F_x) and radial moment (C_x).

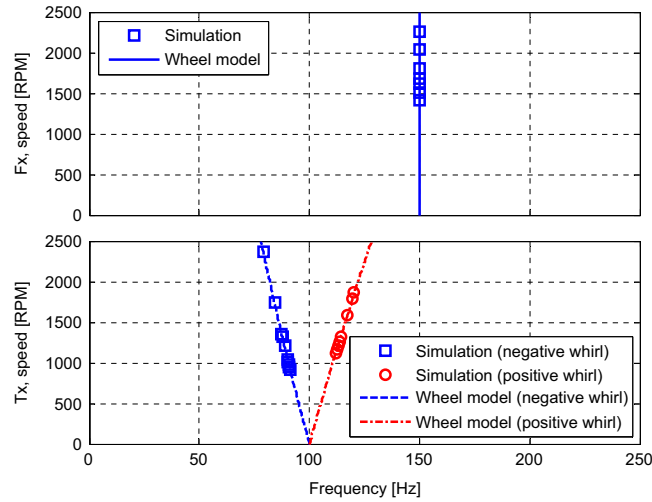


Fig. 8. Estimated natural frequencies of lateral mode (F_x) and whirl mode (T_x).

anti-aliasing filters (KROHN-HITE/3384). The measured wheel speed and disturbance data are shown in Figs. 12 and 13. In the present study, for reasons of wheel information security, the transmitted disturbance forces ($\bar{F}_x$, $\bar{F}_y$, $\bar{F}_z$) and moments ($\bar{T}_x$, $\bar{T}_y$, $\bar{T}_z$) were normalized using the maximum force and moment values, respectively.

4.2. Extended micro-vibration model

The micro-vibration characteristics of the RWA, as shown in Fig. 13, demonstrate that the axial moment ($\bar{T}_z$) was relatively small and negligible compared with the other disturbances; however, the axial force ($\bar{F}_z$) cannot be neglected because it could generate the largest disturbance in other cases. Therefore, in order to establish a practical reaction wheel model, the axial motion should be considered in the equation of motion of the wheel. In the present study, the general equations of motion expressed in Eqs. (3) and (4) were extended in order to include the axial motion along the z-axis. Moreover, in order to represent the non-symmetrical material properties of the flywheel, the parameter values were individually expressed according to each dof. The extended equation of motion can be written as follows:

$$\begin{bmatrix} m_t & 0 & 0 & 0 & 0 \\ 0 & m_t & 0 & 0 & 0 \\ 0 & 0 & m_t & 0 & 0 \\ 0 & 0 & 0 & I_{rrx} & \frac{1}{2} \tilde{I} \sin(2\Omega t) \\ 0 & 0 & 0 & \frac{1}{2} \tilde{I} \sin(2\Omega t) & I_{rry} \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \\ \ddot{\theta} \\ \ddot{\phi} \end{Bmatrix}$$

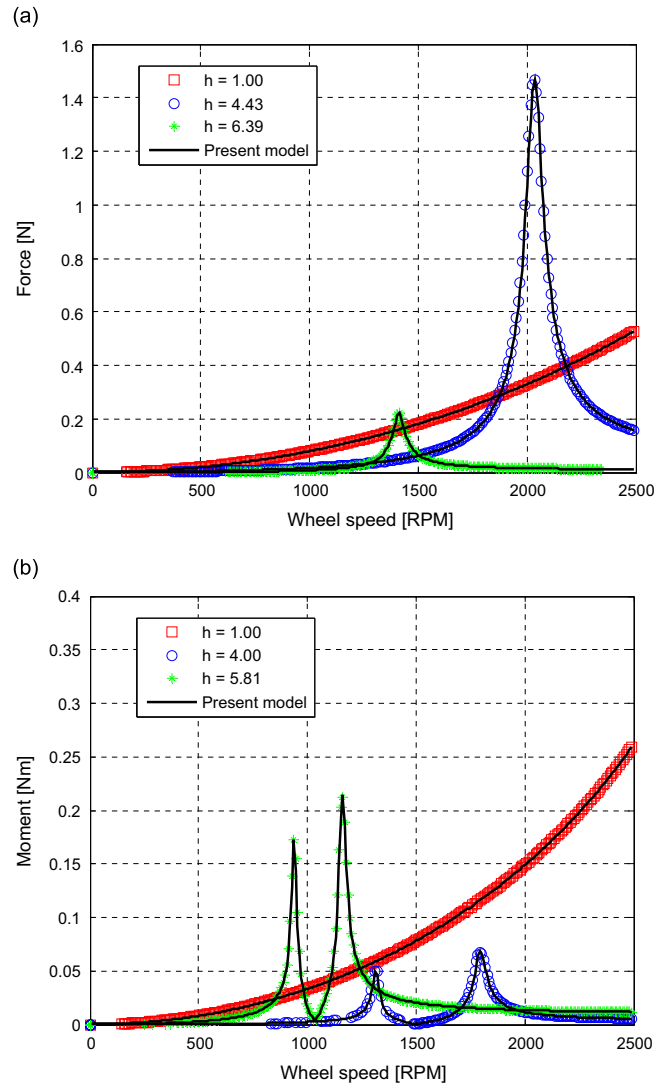


Fig. 9. Estimated harmonic disturbances. (a) Radial force and (b) radial moment.

$$\begin{aligned}
 & + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\Omega \tilde{I} \sin(2\Omega t) & \Omega I_{zz} \\ 0 & 0 & 0 & -\Omega I_{zz} & -\Omega \tilde{I} \sin(2\Omega t) \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\theta} \\ \dot{\phi} \end{Bmatrix} + \begin{bmatrix} c_x & 0 & 0 & 0 & 0 \\ 0 & c_y & 0 & 0 & 0 \\ 0 & 0 & c_z & 0 & 0 \\ 0 & 0 & 0 & c_\theta & 0 \\ 0 & 0 & 0 & 0 & c_\phi \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\theta} \\ \dot{\phi} \end{Bmatrix} \\
 & + \begin{bmatrix} k_x & 0 & 0 & 0 & 0 \\ 0 & k_y & 0 & 0 & 0 \\ 0 & 0 & k_z & 0 & 0 \\ 0 & 0 & 0 & k_\theta & 0 \\ 0 & 0 & 0 & 0 & k_\phi \end{bmatrix} \begin{Bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{Bmatrix} = \begin{Bmatrix} F_x \\ F_y \\ F_z \\ T_x \\ T_y \end{Bmatrix} \quad (12)
 \end{aligned}$$

where c_z , k_z , and F_z are the additional parameters of damping coefficient, stiffness, and disturbance force along the z -axis, respectively.

The coupled micro-vibration model is expressed in the reference frame located at the flywheel CoG. In most practical applications, however, the micro-vibration disturbance cannot be measured at the CoG. As shown in Fig. 11, there are vertical and/or horizontal offsets between the measurement sensor and the flywheel CoG due to the unavoidable interface condition, and the offsets produce an undesired coupling effect of force disturbances in the transmitted moment

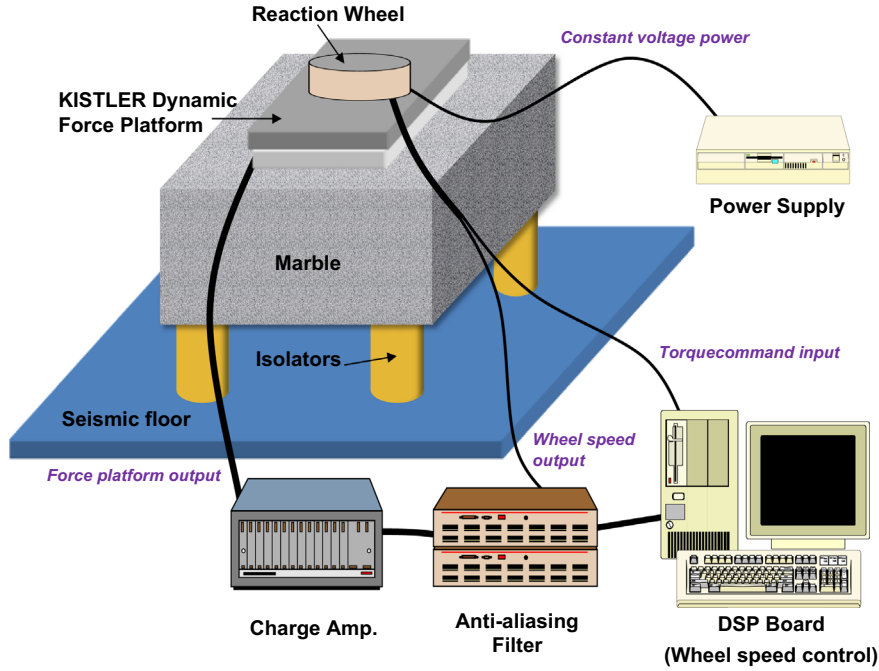


Fig. 10. RWA micro-vibration test setup.

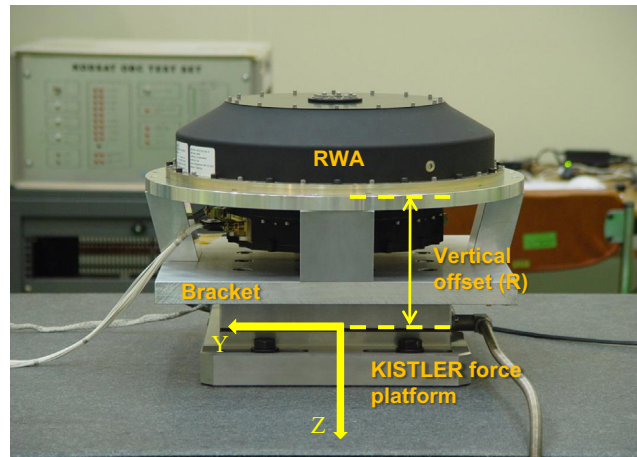


Fig. 11. ITHACO E-type RWA on Kistler platform.

disturbances measured at the sensor plane. Therefore, the equation of motion should be modified in order to consider the geometric offset effects.

The transformation matrix (**B**) from the RWA CoG to the representative measurement point can be expressed as follows:

$$\mathbf{B} = \begin{bmatrix} \mathbf{I}_{3 \times 3} & \begin{bmatrix} 0 & dz \\ -dz & 0 \\ dy & -dx \end{bmatrix} \\ \mathbf{0}_{2 \times 3} & \mathbf{I}_{2 \times 2} \end{bmatrix}, \quad (13)$$

where **I** and **0** are the identity and zero matrices, dx and dy are the horizontal offsets along the x and y directions, respectively, and dz is the vertical offset along the z direction. Using the transformation matrix (**B**), the micro-vibration model written in Eq. (12) can be expressed with respect to the sensor plane as follows:

$$\mathbf{B}^T \mathbf{M}_{\text{total}} \ddot{\mathbf{B}}\mathbf{u} + \mathbf{B}^T \mathbf{C}_{\text{total}} \dot{\mathbf{B}}\mathbf{u} + \mathbf{B}^T \mathbf{K}_{\text{total}} \mathbf{B}\mathbf{u} = \mathbf{B}^T \mathbf{F}, \quad (14)$$

where **u** is the displacement vector, **F** is the disturbance vector, and **M**_{total} and **K**_{total} are the total mass and stiffness matrices of Eq. (12), respectively. The total damping matrix (**C**_{total}) is the sum of the second part (**C**_{gyro}) and third part (**C**) of Eq. (12).

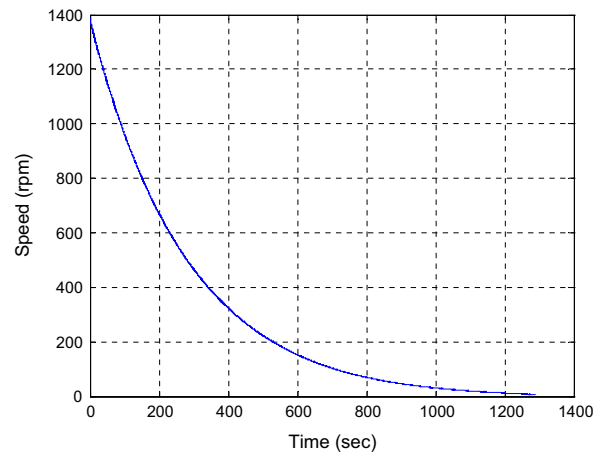


Fig. 12. Wheel speed data measured in free run-down test.

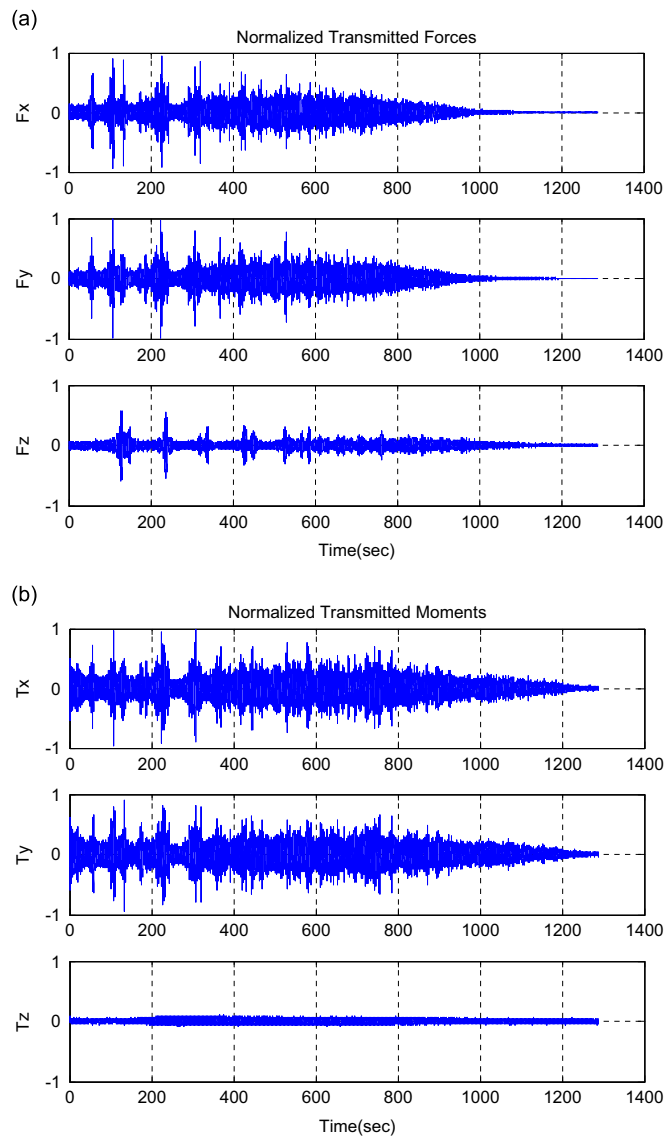


Fig. 13. Normalized transmitted disturbances. (a) Forces and (b) moment.

4.3. Parameter estimation and results

In order to establish the micro-vibration model of the EM wheel, the parameter estimation process was followed using Eq. (14) in the same manner as used in the previous numerical application. In this modeling, the horizontal offsets (dx and dy) were assumed to be zero as a result of the symmetric configuration of the test setup, and the vertical offset (dz) was included in the unknown parameters, which should be determined using Step 4 in the parameter estimation process.

First, using the transmitted disturbance data, the waterfall data were calculated as shown in Fig. 14 (left plots); then, the main harmonic disturbances were selected (right plots). From this harmonic order selection, as listed in Table 3, sixteen main harmonic orders were selected from $h_1^{xy} = 5.23$ to $h_{16}^{xy} = 29.46$ for the radial forces ($\bar{F}_x$ and $\bar{F}_y$), eight orders from $h_1^z = 4.50$ to $h_8^z = 19.34$ for the axial force ($\bar{F}_z$), and fifteen orders from $h_1^{xy} = 1.01$ to $h_{15}^{xy} = 19.75$ for the radial moments ($\bar{T}_x$ and $\bar{T}_y$). Note that the dominant disturbances in the radial and axial force data were not generated by the fundamental harmonic but by the amplification of the higher harmonics due to the structural resonance effect of the flywheel. In addition, the lateral modes (143 Hz) were found not only in the transmitted radial force, but also in the transmitted radial moment (Fig. 14(d) and (f)); this result demonstrates the evidence of the vertical offset effect. Second, the stiffness parameters (k_x , k_y , and k_z) were determined using the parameter optimization process through the application of the translation mode equations expressed in the same form as Eq. (7) to the lateral and axial mode data, respectively. Third, the

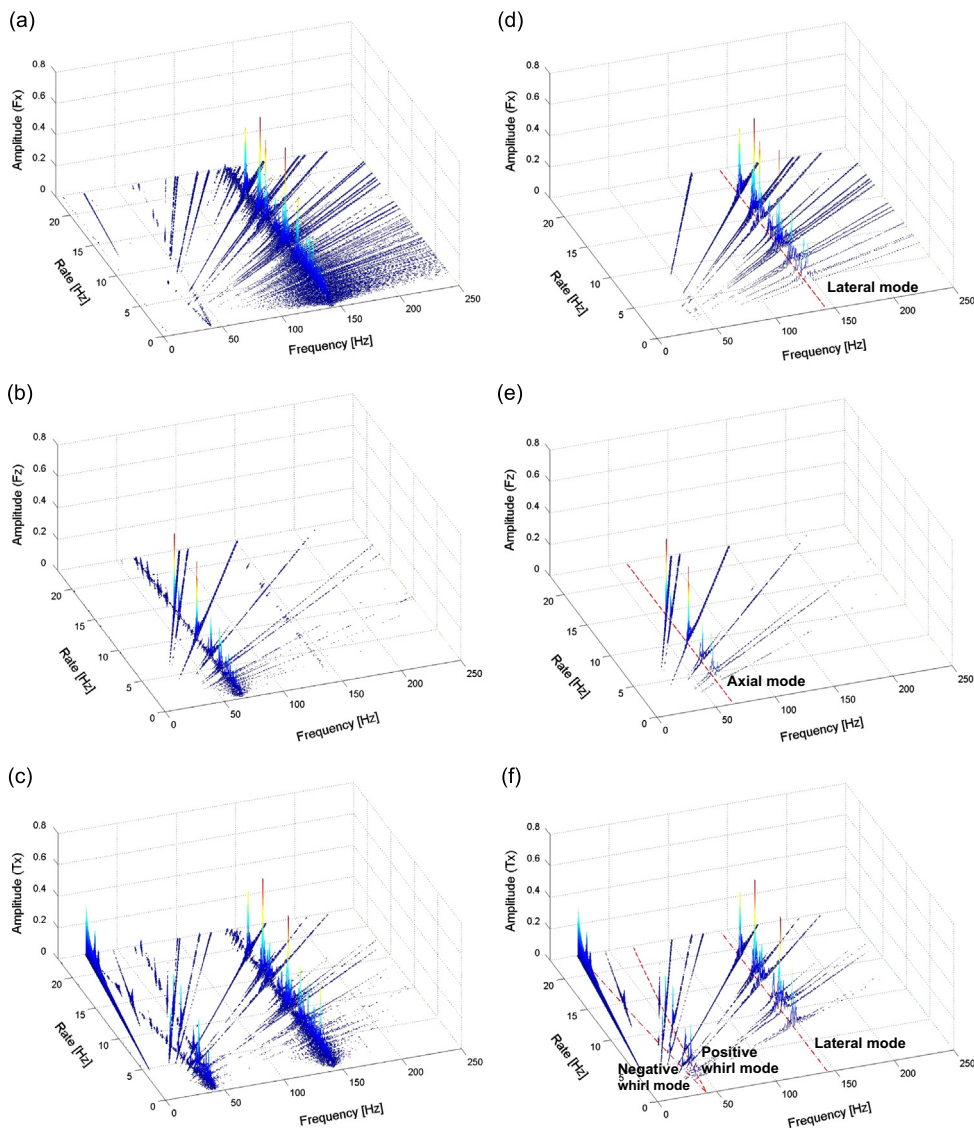


Fig. 14. Waterfall plots of transmitted disturbances. (a) Measured radial force, (b) measured axial force, (c) measured radial moment, (d) selected radial force, (e) selected axial force and (f) selected radial moment.

Table 3
Selected harmonic orders of the EM wheel.

Major harmonic number	Radial force $h_i^{f_{xy}}$	Axial force $h_i^{f_z}$	Radial moment $h_i^{t_{xy}}$
1	5.23	4.5	1.01
2	7.68	4.82	1.19
3	9.01	6.67	2.02
4	9.24	9.04	4.5
5	10.07	9.66	5.24
6	11.67	13.4	5.66
7	13.52	14.52	7.69
8	14.07	19.34	9.01
9	14.36		9.24
10	14.9		10.07
11	18.03		13.54
12	18.95		14.1
13	19.73		14.92
14	23.76		18.95
15	28.63		19.75
16	29.46		

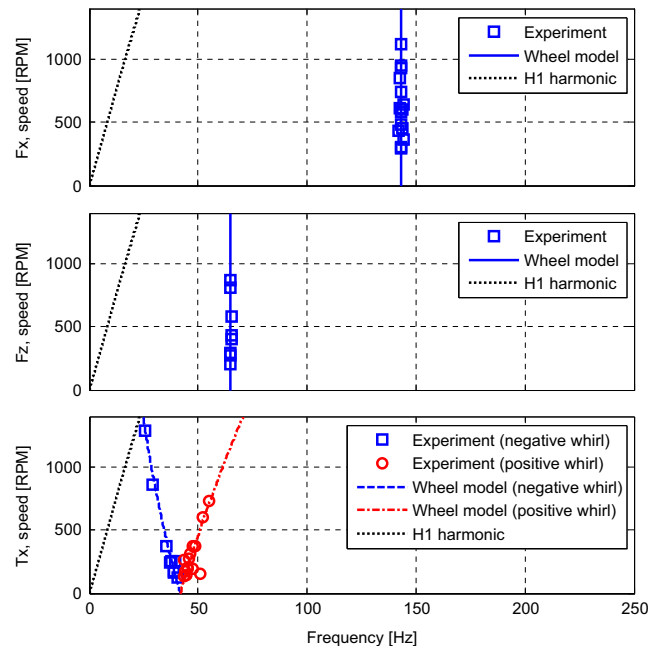


Fig. 15. Frequency identification data of lateral, axial, and whirl modes.

torsional stiffness (k_θ and k_ϕ) and radial MOIs (I_{rx} and I_{ry}) were determined by applying the equations of the torsional mode, i.e. Eq. (8), to the whirl mode data.

Fig. 15 presents the comparison between the numerical mode results and the experimental mode data. The results clearly confirm the good agreement of the estimated wheel model with the modal data. Finally, the damping coefficients (c_x , c_y , c_z , c_θ , and c_ϕ), the harmonic amplitudes ($C_i^{f_x}$, $C_i^{f_y}$, $C_i^{f_z}$, $C_i^{t_x}$, and $C_i^{t_y}$), and the vertical offset (dz) were estimated using the parameter optimization process with the cost function expressed by considering the transformation matrix (**B**).

Fig. 16 presents the comparison between the transmitted disturbance results calculated using the micro-vibration model and the measured disturbance data. On average, the relative errors of the disturbances to the maximum harmonic peaks were 1.03 percent, 1.21 percent, and 1.37 percent for the transmitted disturbance forces ($\bar{F}_x$, $\bar{F}_y$, and $\bar{F}_z$, respectively), and 3.35 percent and 2.69 percent for the disturbance moments ($\bar{T}_x$ and $\bar{T}_y$, respectively). These results indicate that the micro-vibration modeling and parameter estimation methods are sufficient to represent the disturbance characteristics of the wheel.

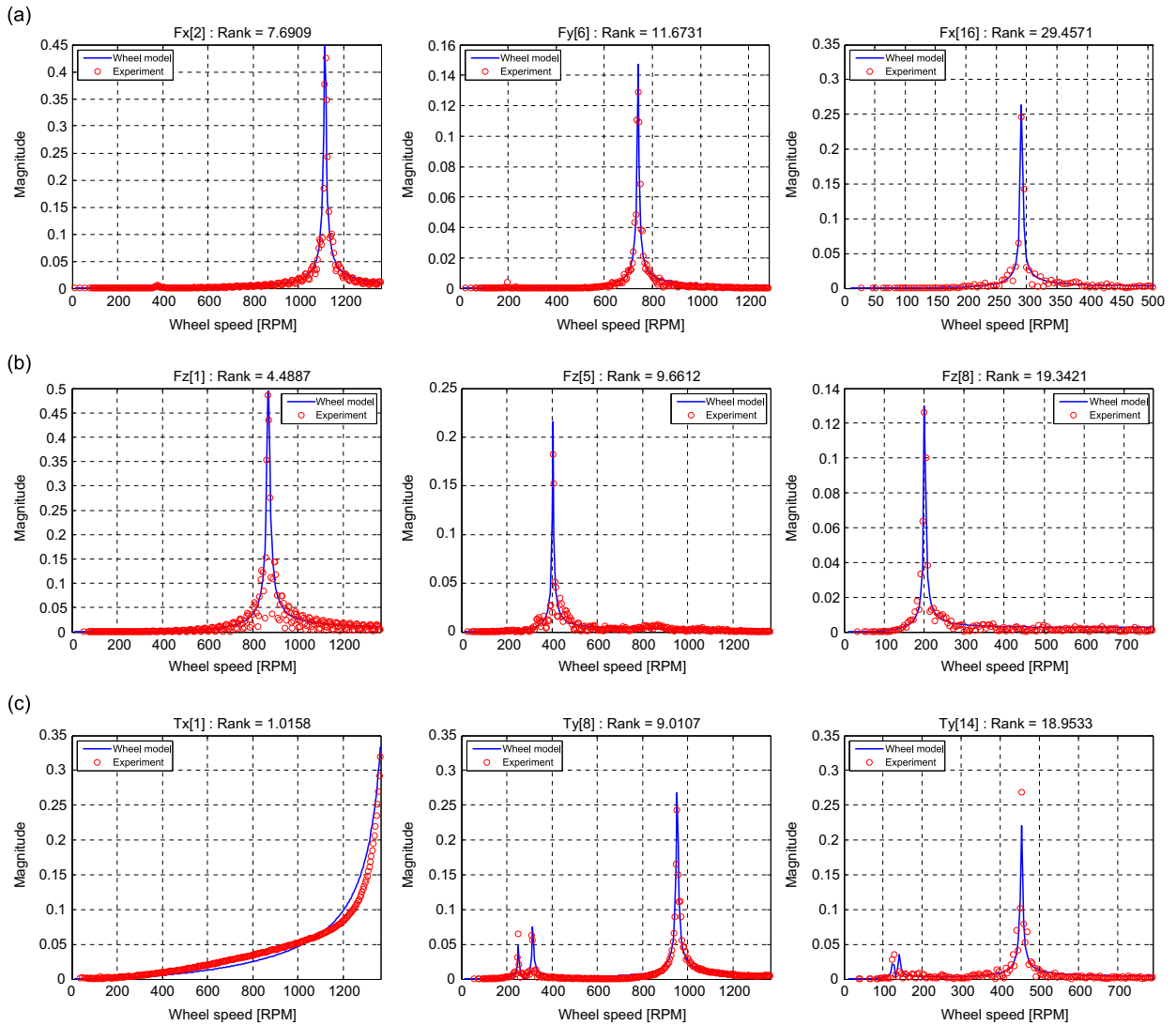


Fig. 16. Harmonic disturbance data. (a) Radial forces (F_x and F_y), (b) axial force (F_z) and (c) radial moments (T_x and T_y).

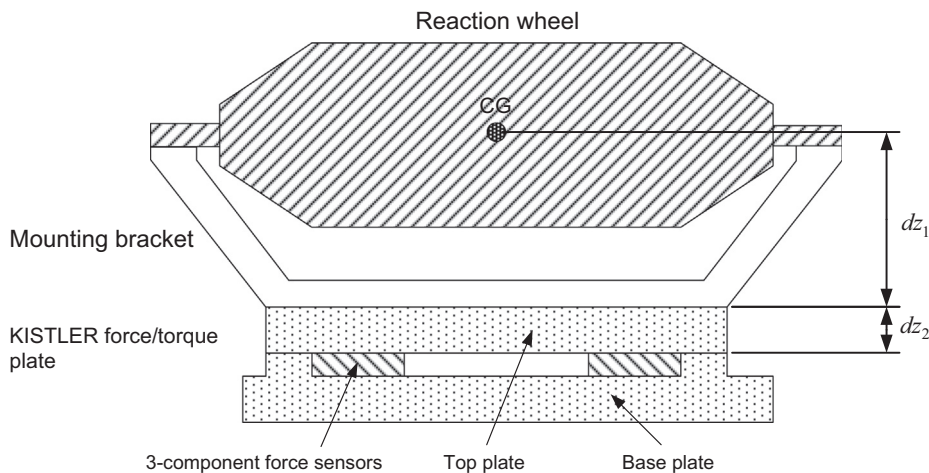


Fig. 17. Cross section of RWA and the Kistler plate.

Note also that the vertical offset was caused by the inescapable interface condition of the mounting bracket, and it is usually determined by measuring the distance (dz_1) from the sensor plane to the flywheel CoG, as shown in Fig. 17. In the present study, the measured offset was 0.145 m. However, the vertical offset value (dz) estimated using Step 4 of the parameter estimation process was 0.185 m and there was a remaining offset distance of 0.04 m. This offset difference could be found through investigating the sensor location of the Kistler plate. As represented in Fig. 17, the plate consists of four 3-component force sensors sandwiched under a high preload between the base and top plates, and the remaining offset distance corresponds with the thickness (dz_2) of the top plate. This result indicates that the vertical offset should be measured from the internal force sensor to the flywheel CoG ($dz=dz_1+dz_2$), and it is additional evidence that demonstrates the validity of the micro-vibration modeling and parameter estimation methods proposed in this study.

5. Conclusions

The micro-vibration model for RWAs is proposed by combining the analytical wheel model and empirical disturbance model. In particular, the parameter estimation process is proposed for the efficient extraction of the parameter values of the micro-vibration model from the disturbance test data. In order to validate the modeling and estimation methods, the micro-vibration model of a numerical RWA was established. The estimated parameter values were compared with the numerical input data, and it was found that the modeling error was less than 2.4 percent. Then, the micro-vibration modeling method was extended to consider the axial motion and the vertical and horizontal offset effects. Finally, the micro-vibration model for the ITHACO E-type wheel was established using the proposed modeling and estimation methods, and the results were compared with the experimental data. From the experimental application, it was confirmed that the micro-vibration modeling and parameter estimation methods are effective for the dynamic disturbance modeling of RWAs. The modeling and estimation methods proposed in the present study are expected to be used effectively in various dynamic problems of RWAs such as wheel isolator design and satellite jitter analyses.

References

- [1] K.B. Doyle, V.L. Genberg, G.J. Michels, *Integrated Optomechanical Analysis*, SPIE, Washington, 2002.
- [2] Gerald C. Holst, *Electro-Optical Imaging System Performance*, 3rd Edition, JCD, Florida, 2002.
- [3] G.W. Neat, J.W. Melody, B.J. Lurie, Vibration attenuation approach for spaceborne optical interferometers, *IEEE Transactions on Control System Technology* 6 (6) (1998) 689–700.
- [4] K.J. Pendergast, C.J. Schauwecker, Use of a passive reaction wheel jitter isolation to meet the Advanced X-ray astrophysics facility imaging performance requirements, *Proceedings of SPIE Astronomical Telescopes and Instruments Conference, Kona, Hawaii* (1998) 1078–1094. (20–28 March).
- [5] K.C. Liu, T. Kenney, P. Maghami, P. Mule, Jitter test program and on-orbit mitigation strategies for solar dynamic observatory, *Proceedings of the 20th International Symposium on Space Flight Mechanics*, NASA Goddard Space Flight Center, September 2007.
- [6] S.E. Miller, P. Kirchman, J. Sudey, Reaction wheel operational impacts on the GOES-N jitter environment, *AIAA Guidance, Navigation and Control Conference and Exhibit*, Hilton Head, South Carolina, AIAA 2007-6736, 20–23 August, 2007, pp. 1–12.
- [7] C. Blaurock, K.C. Liu, P. Mule, Solar dynamic observatory (SDO) HGAS induced jitter, *Proceedings of the 49th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Material Conference*, Schaumburg, Illinois, 7–10 April, 2008, pp. 3140–4149.
- [8] K.C. Liu, P. Maghami, C. Blaurock, Reaction wheel disturbance modeling, jitter analysis, and validation tests for solar dynamics observatory, *AIAA Guidance, Navigation and Control Conference and Exhibit*, Honolulu, Hawaii, AIAA 2008-7232, 18–21 August, 2008, pp. 1–18.
- [9] H.L. Gutierrez, D.W. Miller, *Performance Assessment and Enhancement of Precision Controlled Structures During Conceptual Design*, PhD Thesis, Department of Aeronautics and Astronautics, Massachusetts Institute of Technology, 1999.
- [10] D. Kamesh, R. Pandiyan, A. Ghosal, Modeling, design and analysis of low frequency platform for attenuating micro-vibration in spacecraft, *Journal of Sound and Vibration* 329 (17) (2010) 3431–4350.
- [11] Bill Bialke, High fidelity mathematical modeling of reaction wheel performance, *Advances in the Astronautical Science* (1998) 483–496. (AAS 98-063).
- [12] R.A. Masterson, D.W. Miller, *Development and Validation of Empirical and Analytical Reaction Wheel Disturbance Models*, MS Thesis, Department of Mechanical Engineering, Massachusetts Institute of Technology, 1999.
- [13] B. Li, G. Goddu, M.Y. Chow, Detection of common motor bearing faults using frequency-domain vibration signals and a neural network based approach, *Proceedings of the American Control Conference, Philadelphia, Pennsylvania* (1998) 2032–2036. (June).
- [14] Bill Bialke, A compilation of reaction wheel induced spacecraft disturbances, *Proceedings of the 20th Annual American Aeronautical Society Guidance and Control Conference*, AAS Paper 97-038, February 1997.
- [15] R.A. Masterson, D.W. Miller, R.L. Grogan, Development of empirical and analytical reaction wheel disturbance models, *Proceedings of the 40th AIAA/ASCE/AHS/ASC Structures, Structural Dynamics and Material Conference*, St. Louis, Missouri, 12–15 April, 1999, pp. 1–12.
- [16] R.A. Masterson, D.W. Miller, R.L. Grogan, Development and validation of reaction wheel disturbance models: empirical model, *Journal of Sound and Vibration* 249 (3) (2002) 575–598.
- [17] S. Taniwaki, Y. Ohkami, Experimental and numerical analysis of reaction wheel disturbances, *JSME International Journal Series C* 46 (2) (2003) 519–526.
- [18] H.S. Oh, D.I. Cheon, Precision measurements of reaction wheel disturbances with frequency compensation process, *Journal of Mechanical Science and Technology* 19 (1) (2005) 136–143.
- [19] Y. Zhao, J. Sun, H. Tian, Development of methods identifying parameters in reaction wheel assembly disturbance model, *Aircraft Engineering and Aerospace Technology* 78 (4) (2006) 326–330.
- [20] Z. Zhang, G.S. Alietti, W. Zhou, Microvibrations induced by a cantilevered wheel assembly with a soft-suspension system, *AIAA Journal* 49 (5) (2011) 1067–1079.
- [21] Z. Zhang, G.S. Alietti, W. Ren, Microvibration model development and validation of a cantilevered reaction wheel assembly, *Applied Mechanics and Materials* 226–228 (2012) 133–137.
- [22] W. Zhou, G.S. Aglietti, Z. Zhang, Modelling and testing of a soft suspension design for a reaction/momentum wheel assembly, *Journal of Sound and Vibration* 330 (2011) 4596–4610.
- [23] W. Zhou, D. Li, Design and analysis of an intelligent vibration isolation platform for reaction/momentum wheel assemblies, *Journal of Sound and Vibration* 331 (2012) 2984–3005.
- [24] L.M. Elias, D.W. Miller, A coupled disturbance analysis method using dynamic mass measurement techniques, *Proceedings of the 43rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Material Conference*, Denver, Colorado, 22–25 April 2002, pp. 1–12.

- [25] L.M. Elias, F. Dekens, I. Basdogan, L. Sievers, T. Neville, A methodology for “modeling” the mechanical interaction between a reaction wheel and a flexible structure, *Proceedings of SPIE Astronomical Telescope and Instrumentation Conference*, Waikoloa, Hawaii, 22–28 August 2002.
- [26] F.C. Lee, M. Werner, Reaction wheel jitter analysis including rocking dynamics & bearing harmonic disturbances, *Advances in the Astronautical Science*, AAS 07-006 (2007) 93–110.
- [27] Jan J. Tuma, *Engineering Mathematics Handbook*, 4th Edition, McGraw-Hill, New York, 1997.
- [28] J.C. Lagarias, J.A. Reeds, M.H. Wright, P.E. Wright, Convergence properties of the Nelder–Mead simplex method in low dimensions, *SIAM Journal of Optimization* 9 (1) (1998) 112–147.